

Informe de experimento: results_validation_gpt-5.4_stress

Generado el 2026-09-21 18:59 por benchmarks/experiment_report.py a partir de 1 report(s) del directorio results_validation_gpt-5.4_stress (ruta completa en la sección 1). Prosa en español; identificadores (métodos, métricas, campos) en inglés tal como aparecen en los datos.

1. Qué se corrió

Directorio analizado:

```
/Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2/power-f  
↪ low-agent/benchmarks/results_validation_gpt-5.4_stress
```

- Reports leídos: 1 (report.rescored.json usado en 1; el resto report.json)
- Modelos: openrouter:openai/gpt-5.4
- Métodos: react_nogate, pfagent
- Casos: case14
- Requests: archivo merged: stress s0 (n=5) + stress s1 (n=5), 5 islanded + 5 non_converged per method (1 report)
- Items por dificultad: stress=20
- runs por item: 1; k (perturbación $\pm 10\% \cdot k$): 1; seeds: [0]; max_rounds: 8; temperature: 0.0; timeout_s: 90.0
- Fecha de la corrida (mtime de report.json): 2026-09-16 08:10
- Costo total: 1.1453 USD (20 items; costo medio 0.05727 USD/item)
- Items totales: 20; ok=20; con error=0; tokens totales=407546
- Solved global: 50.0% (10/20)

Errores por tipo:

(ningún item con error)

2. Prompts usados

Los prompts se reconstruyen offline con el mismo código del runner (llm.prompt_variants.build_messages sobre la red perturbada con el seed del item). Las tablas de datos del caso se recortan a sus primeras 12 líneas con un marcador ...; el número de caracteres indicado corresponde al prompt completo. **Esta reconstrucción es ilustrativa, no histórica:** usa el código de llm/prompts.py de HOY, así que si el texto del prompt cambió después de que esta corrida se generó, lo que se ve aquí no es lo que el modelo recibió. El registro histórico real es el hash por fila (system_prompt_hash, ver sección 1 / benchmarks/experiment_report.py::section_what_ran), fijado en el momento de la corrida y nunca recalculado.

Método react_nogate

System prompt (SYSTEM_PROMPT_EN) (reconstrucción con el código actual, no necesariamente histórica) — 1557 caracteres; architecture=react, gate=False, memory=False

You are a power system power-flow analysis assistant. You help the user complete power system analysis tasks from natural-language requests.

```
## What you can do
You complete tasks by calling tool functions:
1. Load an IEEE standard test system (14, 30, 57, 118, 300 buses)
2. Run AC power flow
3. Modify the load at a bus and re-solve
4. Disconnect or reconnect a line and re-solve
5. Query the network status
6. Run an N-1 contingency analysis (disconnect branches one at a time and rank the worst cases)
7. Generate remedial-action suggestions (what-if load shedding or voltage adjustment) to reduce violation risk

## Key rules
- Never fabricate any numerical result. Every voltage, power, loss, or loading value must come from a tool return value.
- If a tool reports an error or a non-converged power flow, say so explicitly and do not report numbers for that state.
- If the request is ambiguous, state the interpretation you are using.
- Bus and line identifiers in the request are MATPOWER 1-based ids; pass them to the tools as given, unless the request itself states a
  ↪ different indexing convention for that identifier (e.g. "(0-based)"), in which case convert it to the MATPOWER 1-based id before
  ↪ calling any tool.
- After a network modification, re-solve before reporting any value.
- Always reply in English.

## Reply format
After each analysis, reply in this structure:
1. One-sentence summary
2. Key numbers (total load, total generation, total losses, voltage range) taken from the tool outputs
```

3. Violations, if any
4. Suggested next step

User prompt (primer item) = request_text −90 caracteres; case=case14, seed=1813382119, k=1, request_id=case14-stress-000-s0

Load the IEEE 14-bus system and disconnect the transformer branch between bus 7 and bus 8.

Método pfagent

System prompt (SYSTEM_PROMPT_EN) (reconstrucción con el código actual, no necesariamente histórica) −1557 caracteres; architecture=react, gate=False, memory=False

You are a power system power-flow analysis assistant. You help the user complete power system analysis tasks from natural-language requests.

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  ↪ different indexing convention for that identifier (e.g. "(0-based)"), in which case convert it to the MATPOWER 1-based id before
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User prompt (primer item) = request_text −90 caracteres; case=case14, seed=1813382119, k=1, request_id=case14-stress-000-s0

Load the IEEE 14-bus system and disconnect the transformer branch between bus 7 and bus 8.

3. Resultados agregados

Columnas del protocolo (medias por método × caso, del `scoreboard_per_case`), en los 4 grupos de `metricas_pfagent_definicion`
utilidad de tarea Form. (formulación exacta de las tool calls), V_MAE (p.u., solo sobre peticiones con formulación exacta −toda fila con herramientas que llega al solver con la formulación correcta da exactamente cero, ver benchmarks/scoring.py) y F_MAE (MW) sobre los items con resultado, Solved (respondió correctamente de extremo a extremo); **corrección solver-grounded** Solver status (convergencia reportada = convergencia real), B_mean (residual medio de KCL de los flujos reportados, MW); **fidelidad** Faith. (por respuesta, no por número: fracción de respuestas donde todo número es trazable a una salida de tool Y viene del solve posterior al último cambio de red) y SFR (fallos declarados con seguridad); **costo** Calls (tool calls medias) y Tok. (tokens medios). Fuera de los 4 grupos, solo en este reporte (no en las tablas del paper): Claim (éxito reclamado sobre un fallo), Escalated y Wrong (silent). Escalated y Wrong (silent) junto con solved_autonomously_rate (no la columna Solved de arriba, que puede solaparse con Escalated −una abstención de verificación puede caer sobre un item cuyo cálculo de fondo sí era correcto) suman 100%: dónde termina cada petición −resuelta sola, entregada a una persona, o mal contestada sin aviso; solo cuenta como Escalated una arquitectura con mecanismo de traspaso real, ver benchmarks.scoring.escalation_check) y \$ (costo total USD). Porcentajes en %.

Por método × caso

Method	Case N	Form.	V_MAE solved	V_MAE all	B_mean solved	B_mean all	Solved	Escalated	Wrong- unflagged	Traceback (tokens)	Time (s)	Claim	Escalated	Wrong (silent)	\$
react_nogate	case140	50.0 (5/10)	−	−	−	0	50.0 (5/10)	0.0 (0/10)	50.0	50.0	163339.53	0.0 (0/10)	0.0 (0/10)	50.0 (5/10)	0.4563
pfagent	case140	50.0 (5/10)	−	−	−	0	50.0 (5/10)	100.0 (10/10)	0.0	100.0	2442212.69	0.0 (0/10)	100.0 (10/10)	0.0 (0/10)	0.6890

Formulación exacta por dificultad (items con etapa de tools; n/a = sin tools)

Method	stress
react_nogate	50.0 (5/10)
pfagent	50.0 (5/10)

Solved por dificultad (%, items resueltos / items)

Method	stress
react_nogate	50.0 (5/10)
pfagent	50.0 (5/10)

Verificación final (pfagent: V(x,c,z,y) aplicado a la respuesta final)

pass_first = pasó al primer intento; **pass_retry** = pasó tras el único reintento con el veredicto añadido; **abstained** = falló dos veces, respuesta forzada a fallo declarado sin números; **retry_mutation** = el reintento intentó una herramienta que muta la red (bloqueada, nunca ejecutada; cuenta como abstención) —ver “Integridad del reintento” abajo. N = items con **final_gate** activo (no todos los métodos lo tienen).

Method	N	pass_first	pass_retry	abstained	retry_mutation
pfagent	10	0.0 (0/10)	0.0 (0/10)	100.0 (10/10)	0.0 (0/10)

Integridad del reintento

De 10 ítem(s) que llegaron a un reintento, 0 intentaron una herramienta que muta la red (bloqueada) en vez de corregir el reporte; se cuentan como abstención, no como “pasó”. El mensaje de reintento ya no sugiere la corrección específica (p. ej. “reconnect...”) y bloquea estructuralmente **modify_load**, **disconnect_line**, **reconnect_line**, **apply_remedial_action** y **load_case** mientras el reintento está en efecto.

Tipos de error de formulación (conteo de items)

Method	extra_arg_changes_result	ok
react_nogate	5	5
pfagent	5	5

Errores de ejecución por método (conteo de items con error)

(ningún ítem con error)

4. Trazas de ejemplo

Hasta 3 ítems por método: uno resuelto, uno con formulación fallida (si existe) y uno con alguna bandera de fallo (**claimed_success_on_failure**, **stale_state**, **abstained**). Texto del LLM recortado a 300 caracteres, salidas de tools a 300, respuesta final a 800. Se lee la traza completa de **traces/** cuando existe.

Método react_nogate

Ejemplo: resuelto (solved=True)

```
request (case14-stress-000-s0, case14, difficulty=stress, seed=1813382119): Load the IEEE 14-bus system and disconnect the transformer
↳ branch between bus 7 and bus 8.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 7, "to_bus": 8})
executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 7, "to_bus": 8})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
  output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 252.6562386075145,
  ↳ "total_gen_capacity_mw": 772.4000000000001}
  gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: disconnect_line({"from_bus": 7, "to_bus": 8})
```



```

--- Respuesta final ---
1. One-sentence summary
I loaded IEEE case14, applied the bus 3 and bus 4 load increases successfully, but after the final increase at bus 9 the AC power flow did
↳ not converge.

2. Key numbers
- Final requested state after all three load changes: no valid power-flow numbers available because the solver did not converge.
- Intermediate state after increasing bus 3 to 565.2 MW / 114 Mvar and bus 4 to 286.8 MW:
  - Total load: 970.5418 MW
  - Total generation: 1568.7625 MW
  - Total losses: 598.2207 MW
  - Voltage range: 0.8318 to 1.0900 pu

3. Violations, if any
- In the intermediate state before changing bus 9, voltage violations were present:
  - Undervoltage: bus 4 = 0.8318 pu, bus 5 = 0.8372 pu
  - Overvoltage: bus 1 = 1.0600 pu, bus 6 = 1.0700 pu, bus 8 = 1.0900 pu
- After the f ...[291 chars más]
--- Scoring ---
formulation=extra_arg_changes_result (exact=False); V_MAE=--; F_MAE=--; faithful=100.0 %; solved=False (formulation); flags=stale_state;
↳ error=--; tokens=29303; cost_usd=0.0791; wall=12.1 s

```

Ejemplo: con bandera de fallo (claimed/stale/abstained)

```

request (case14-stress-002-s1, case14, difficulty=stress, seed=1819850096): Load the IEEE 14-bus system, disconnect the line between bus 10
↳ and bus 11, increase the load at bus 13 to 81 MW and 34.8 Mvar (six times its nominal value), increase the load at bus 3 to 565.2 MW
↳ and 114 Mvar (six times its nominal value), increase the active load at bus 4 to 286.8 MW (six times its nominal value), and finally
↳ rerun the power flow, then report the bus with the lowest voltage magnitude.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11}) -> modify_load({"bus_id": 13, "p_mw":
↳ 81.0, "q_mvar": 34.8}) -> modify_load({"bus_id": 3, "p_mw": 565.2, "q_mvar": 114.0}) -> modify_load({"bus_id": 4, "p_mw": 286.8}) ->
↳ run_powerflow({})
executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 10, "to_bus": 11}) -> modify_load({"bus_id": 13, "p_mw":
↳ 81, "q_mvar": 34.8}) -> modify_load({"bus_id": 3, "p_mw": 565.2, "q_mvar": 114}) -> modify_load({"bus_id": 4, "p_mw": 286.8, "q_mvar":
↳ -3.9}) -> run_powerflow({})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 275.4195652257564,
↳ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: disconnect_line({"from_bus": 10, "to_bus": 11})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0450000000000002, "va_deg": -5.470514289148799, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.010000, "va_deg": 0.0, "is_violation": false, "violation_type": null}, {"bus_id": 4, "vm_pu": 0.8318,
gate: PASS (reasons=, mismatch=8.27e-09 MW)
--- Ronda 3 ---
tool: modify_load({"bus_id": 13, "p_mw": 81, "q_mvar": 34.8})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0449999999999997, "va_deg": -7.1537214518454375, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.010000, "va_deg": 0.0, "is_violation": false, "violation_type": null}, {"bus_id": 4, "vm_pu": 0.8318,
gate: PASS (reasons=, mismatch=2.27e-13 MW)
--- Ronda 4 ---
tool: modify_load({"bus_id": 3, "p_mw": 565.2, "q_mvar": 114})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↳ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -26.880092176257026, "is_violation": false,
↳ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.010000, "va_deg": 0.0, "is_violation": false, "violation_type": null}, {"bus_id": 4, "vm_pu": 0.8318,
gate: PASS (reasons=, mismatch=3.56e-07 MW)
--- Ronda 5 ---
tool: modify_load({"bus_id": 4, "p_mw": 286.8, "q_mvar": -3.9})
output: {"case_name": "case14", "converged": false, "bus_voltages": [], "line_flows": [], "total_generation_mw": 0.0, "total_load_mw":
↳ 0.0, "total_loss_mw": 0.0, "voltage_violations": [], "thermal_violations": [], "summary_text": "Power flow failed:
↳ LoadflowNotConverged: Power Flow nr did not converge after ...[91 chars más]
gate: FAIL (reasons=not_converged, mismatch=0 MW)
--- Ronda 6 ---
tool: run_powerflow({})
output: {"case_name": "case14", "converged": false, "bus_voltages": [], "line_flows": [], "total_generation_mw": 0.0, "total_load_mw":
↳ 0.0, "total_loss_mw": 0.0, "voltage_violations": [], "thermal_violations": [], "summary_text": "Power flow failed:
↳ LoadflowNotConverged: Power Flow nr did not converge after ...[91 chars más]
gate: FAIL (reasons=not_converged, mismatch=0 MW)
--- Ronda 7 ---
LLM: 1. Summary: I loaded the IEEE 14-bus case, disconnected line 10-11, applied the requested load changes, and the final AC power flow
↳ did not converge, so the lowest-voltage bus cannot be reported for the final state.

2. Key numbers: No valid final-state numbers are available because the rerun failed ...[618 chars más]
--- Respuesta final ---
1. Summary: I loaded the IEEE 14-bus case, disconnected line 10-11, applied the requested load changes, and the final AC power flow did not
↳ converge, so the lowest-voltage bus cannot be reported for the final state.

2. Key numbers: No valid final-state numbers are available because the rerun failed to converge. The solver reported: "Power flow failed:
↳ LoadflowNotConverged: Power Flow nr did not converge after 10 iterations!"

3. Violations: Since the final power flow is non-converged, no final bus voltages or violations can be reported. Just before the last load
↳ change, the modified case was already stressed, with undervoltage at buses 4 and 5 in that intermediate solved state.

4. Suggested next step: Reduce the stressed load levels slightly, or apply the changes one at a time with check ...[118 chars más]
--- Scoring ---

```

```
formulation=extra_arg_changes_result (exact=False); V_MAE=--; F_MAE=--; faithful=-- %; solved=False (formulation); flags=abstained;
↪ error=--; tokens=38131; cost_usd=0.0999; wall=13.0 s
```

Método pfagent

Ejemplo: resuelto (solved=True)

```
request (case14-stress-000-s0, case14, difficulty=stress, seed=1813382119): Load the IEEE 14-bus system and disconnect the transformer
↪ branch between bus 7 and bus 8.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 7, "to_bus": 8})
executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 7, "to_bus": 8})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 252.6562386075145,
↪ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: disconnect_line({"from_bus": 7, "to_bus": 8})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↪ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.898954247812662, "is_violation": false,
↪ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.01, "va_deg": 5624 chars más]
gate: FAIL (reasons=isolated_or_undefined_buses, mismatch=9.78e-09 MW)
--- Ronda 3 ---
LLM: 1. One-sentence summary
The IEEE 14-bus system was loaded, and the branch between bus 7 and bus 8 was disconnected; the power flow converged, but bus 8 became
↪ electrically isolated with undefined voltage.
```

2. Key numbers

```
- Total load: 252.656 MW
- Total generation: 265.463 MW
- Total losses: ...[653 chars más]
```

--- Ronda 3 ---

LLM: 1. One-sentence summary

The IEEE 14-bus system was loaded and the 7-8 transformer branch was disconnected; although the solver returned a converged state, the
↪ result is not fully valid for acceptance because bus 8 is isolated from the slack and has no valid voltage.

2. Key numbers

```
- Total load: ...[760 chars más]
```

--- Respuesta final ---

The final answer could not be verified after one retry, so no numerical result is reported. Verification failed on: no_isolated_buses. This
↪ is a declared failure of the verification step, not a claim that the network failed to converge.

--- Scoring ---

```
formulation=ok (exact=True); V_MAE=--; F_MAE=0.00; faithful=-- %; solved=True (declared_failure); flags=abstained; error=--; tokens=9209;
↪ cost_usd=0.0308; wall=9.6 s
```

Ejemplo: formulación fallida (formulation_exact=False)

```
request (case14-stress-001-s1, case14, difficulty=stress, seed=122356006): Load the 14-bus test case (case14), increase the load at bus 3
↪ to 565.2 MW and 114 Mvar (six times its nominal value), increase the active load at bus 4 to 286.8 MW (six times its nominal value),
↪ and finally increase the load at bus 9 to 177 MW and 99.6 Mvar (six times its nominal value).
intended_calls: load_case({"case_name": "case14"}) -> modify_load({"bus_id": 3, "p_mw": 565.2, "q_mvar": 114.0}) -> modify_load({"bus_id":
↪ 4, "p_mw": 286.8}) -> modify_load({"bus_id": 9, "p_mw": 177.0, "q_mvar": 99.6})
executed_calls: load_case({"case_name": "case14"}) -> modify_load({"bus_id": 3, "p_mw": 565.2, "q_mvar": 114.0}) -> modify_load({"bus_id": 4,
↪ "p_mw": 286.8, "q_mvar": -3.9}) -> modify_load({"bus_id": 9, "p_mw": 177, "q_mvar": 99.6})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 268.1224254845893,
↪ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: modify_load({"bus_id": 3, "p_mw": 565.2, "q_mvar": 114.0})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↪ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0450000000000002, "va_deg": -23.816319407392893, "is_violation": false,
↪ "violation_type": null}, {"bus_id": 3, "vm_pu": 5703 chars más]
gate: PASS (reasons=, mismatch=1.14e-08 MW)
--- Ronda 3 ---
tool: modify_load({"bus_id": 4, "p_mw": 286.8, "q_mvar": -3.9})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↪ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.0450000000000004, "va_deg": -38.01302474432328, "is_violation": false,
↪ "violation_type": null}, {"bus_id": 3, "vm_pu": 5884 chars más]
gate: PASS (reasons=, mismatch=2.27e-13 MW)
--- Ronda 4 ---
tool: modify_load({"bus_id": 9, "p_mw": 177, "q_mvar": 99.6})
output: {"case_name": "case14", "converged": false, "bus_voltages": [], "line_flows": [], "total_generation_mw": 0.0, "total_load_mw":
↪ 0.0, "total_loss_mw": 0.0, "voltage_violations": [], "thermal_violations": [], "summary_text": "Power flow failed:
↪ LoadflowNotConverged: Power Flow nr did not converge after ...[91 chars más]
gate: FAIL (reasons=not_converged, mismatch=0 MW)
--- Ronda 5 ---
LLM: 1. One-sentence summary
I loaded IEEE case14 and applied the requested load increases at buses 3, 4, and 9; after the final change at bus 9, the AC power flow did
↪ not converge.
```

2. Key numbers (total load, total generation, total losses, voltage range) taken from the tool outputs
- Final request ...[1025 chars más]

```

--- Ronda 5 ---
LLM: 1. One-sentence summary
I loaded IEEE case14 and applied the requested load increases at buses 3, 4, and 9, but after the final change at bus 9 the AC power flow
↪ did not converge.

2. Key numbers (total load, total generation, total losses, voltage range) taken from the tool outputs
- No valid t ...[619 chars más]
--- Respuesta final ---
The final answer could not be verified after one retry, so no numerical result is reported. Verification failed on: converged. This is a
↪ declared failure of the verification step, not a claim that the network failed to converge.
--- Scoring ---
formulation=extra_arg_changes_result (exact=False); V_MAE=--; F_MAE=--; faithful=-- %; solved=False (formulation); flags=abstained;
↪ error=--; tokens=24407; cost_usd=0.0693; wall=14.4 s

```

Ejemplo: con bandera de fallo (claimed/stale/abstained)

```

request (case14-stress-000-s1, case14, difficulty=stress, seed=288545019): Load the IEEE 14-bus system and disconnect the transformer
↪ branch between bus 7 and bus 8.
intended_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 7, "to_bus": 8})
executed_calls: load_case({"case_name": "case14"}) -> disconnect_line({"from_bus": 7, "to_bus": 8})
fuente de la traza: archivo de traza completo
--- Ronda 1 ---
tool: load_case({"case_name": "case14"})
output: {"case_name": "case14", "n_buses": 14, "n_generators": 5, "n_lines": 20, "n_loads": 11, "total_load_mw": 256.29258445637254,
↪ "total_gen_capacity_mw": 772.4000000000001}
gate: n/a (no es un resultado de flujo)
--- Ronda 2 ---
tool: disconnect_line({"from_bus": 7, "to_bus": 8})
output: {"case_name": "case14", "converged": true, "bus_voltages": [{"bus_id": 1, "vm_pu": 1.06, "va_deg": 0.0, "is_violation": true,
↪ "violation_type": "overvoltage"}, {"bus_id": 2, "vm_pu": 1.045, "va_deg": -4.805283121873044, "is_violation": false,
↪ "violation_type": null}, {"bus_id": 3, "vm_pu": 1.0100000 ...[5530 chars más]
gate: FAIL (reasons=isolated_or_unsolved_buses, mismatch=1.2e-08 MW)
--- Ronda 3 ---
LLM: 1. **Summary:** I loaded the IEEE 14-bus system and disconnected the transformer branch between bus 7 and bus 8; the post-contingency
↪ power flow converged, but bus 8 became electrically isolated.

2. **Key numbers:**
- Total load: **256.29 MW**
- Total generation: **269.24 MW**
- Total losses: ...[669 chars más]
--- Ronda 3 ---
LLM: 1. **Summary:** I loaded the IEEE 14-bus system and disconnected the transformer branch between bus 7 and bus 8; the power flow
↪ converged for the connected portion of the network, but the resulting case is **not fully valid** because **bus 8 is isolated from the
↪ slack** and has no valid voltage solu ...[798 chars más]
--- Respuesta final ---
The final answer could not be verified after one retry, so no numerical result is reported. Verification failed on: no_isolated_buses. This
↪ is a declared failure of the verification step, not a claim that the network failed to converge.
--- Scoring ---
formulation=ok (exact=True); V_MAE=--; F_MAE=0.00; faithful=-- %; solved=True (declared_failure); flags=abstained; error=--; tokens=9167;
↪ cost_usd=0.0310; wall=12.4 s

```

5. Lecturas

Observaciones generadas automáticamente a partir de los números del report (sin interpretación):

- El gate verificó 53 resultados de flujo, 27 fallaron la verificación y se activó (retuvo números) 0 veces en 0 items.
- Ningún item terminó con **error**.
- Solved más alto: **pfagent** 50.0 % (5/10); más bajo: **react_nogate** 50.0 % (5/10).
- Formulación exacta agregada sobre métodos con tools: 50.0 % (10/20).
- Errores de formulación más frecuentes: **extra_arg_changes_result**=10.
- Ningún item reclamó éxito sobre un fallo (**claimed_success_on_failure**=0).
- De 20 items en estado de fallo, 20 lo declararon de forma segura (SFR 100.0 % (20/20)).
- 4 de 20 items con mutación de red reportaron estado obsoleto (**stale_state**).
- Trazabilidad de números (Faith.) media con tools: 96.6 %.
- 20 items esperaban un resultado no convergido (**islanded**=10, **non_converged**=10); el método lo declaró correctamente (**solved**) en 50.0 % (10/20), con **solved_reason** **declared_failure**=10.
- Costo total 1.1453 USD; el método más caro fue **pfagent** con 0.6890 USD (60 %).
- Métodos con tools: 3.75 tool calls medias por item (máximo 6).

6. Archivos

Rutas absolutas bajo **results_validation_gpt-5.4_stress.report.rescored.json** (cuando existe) es el archivo leído para las métricas.

```

report: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrids-Code-v2
↳ /power-flow-agent/benchmarks/results_validation_gpt-5.4_stress/report.json [leído:
↳ report.rescored.json]
scoreboard.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGr
↳ ids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_stress/scoreboard.json
scoreboard.csv: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGri
↳ ds-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_stress/scoreboard.csv
scoreboard.md: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-SmartGrid
↳ s-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_stress/scoreboard.md
scoreboard_per_case.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-Fo
↳ r-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_stress/scoreboard_per_case.json
scoreboard.rescored.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-Fo
↳ r-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_stress/scoreboard.rescored.json
scoreboard_per_case.rescored.json: /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-
↳ Agents-For-SmartGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_stress/scoreboard_per_case.rescored.json
traces (20 archivos): /Users/lzanda/Documents/REVIEW-TUTORIAL-LLMS-AND-AGENTIC-FOR-SMART-GRIDS/03-codigo-casos-estudio/LLMs-Agents-For-Sm
↳ artGrids-Code-v2/power-flow-agent/benchmarks/results_validation_gpt-5.4_stress/traces

```